CAO THANH TUNG

+84 356454985 | cs.tungthanh@gmail.com

linkedin.com/in/tungthanh14
github.com/cs-tungthanh

OBJECTIVE

I'm a passionate engineer with 6 years of experience specializing in distributed systems, IoT backends, cloud infrastructure on AWS, and performance optimization. Led teams of up to 6 engineers, delivering production robot fleet management systems from real-time monitoring to state machine-based task executors. Passionate about clean architecture, system reliability, and building things that scale.

SKILL

- **Programming Languages:** Golang, NodeJS, Python.
- **Databases & Storage**: PostgreSQL, DynamoDB, Redis, S3.
- **Others**: RESTful API, AWS Services, SOLID, Hexagonal Architecture, Event-Driven Architecture, Message Queues, ...

EXPERIENCE

Software Engineer **ROBO-HI (ZMP) VIETNAM CO. LTD** **Nov 2020 – Now**

- **Robot Backend Service and Cloud Infrastructure (2022 - Present)**

  Lead a team of 4~6 members building core robot services for tracking, control, and analytics, spanning both **on-robot client agents** and **cloud backend services** to manage autonomous robots at scale.

  Contribute to delivering 2 major projects in the company:
  - Warehouse project: Developed and controlled forklift robots for automated pick-and-place operations and seamless integration with the Autolator system.
  - Takanawa Gateway Project: a smart city robotics platform integrating delivery, surveillance, and cleaning robots for public facilities, enabling multi-robot coordination and monitoring.

  Robot Client Agent (NodeJS, Python, MQTT, Redis PubSub)
  - Designed and built from scratch, Agent Client is installed on robots and manages bidirectional communication with the backend system via AWS IoT Core (MQTT) and REST APIs, applying Hexagonal Architecture to fully decouple business logic from infrastructure.
  - Designed dual deployment models (on-device and server-side) supporting both SDK and Docker packaging, enabling flexible integration across multiple robot vendors.
  - Designed and built simulation tools for **integration testing** and **24/7 client** demonstrations, significantly improving system reliability and stability.

  Tracking and Analytic Service (Golang, AWS Services)
  - Design and build a real-time telemetry pipeline (Kinesis + Lambda) for ordered per-robot processing and horizontal scalability.
  - Designed a tiered storage strategy separating static equipment/robot data (IDs, configs, ...) in PostgreSQL from high-frequency dynamic state in DynamoDB, reducing PostgreSQL load by **over 80%** and improving overall system throughput.
  - Archive telemetry data to S3 with Athena-based analytics, reducing storage costs and enabling ad-hoc operational queries.
  - Built a visualization and analytics layer, generating event timelines and images to analyze robot behavior, combining telemetry and cross-service logs to support debugging and performance validation.
  - Built Glue ETL jobs for data enrichment and migration.

  Traffic Management (NodeJS, Dynamo, Lambda)
  - Implemented **priority-zone-based flow control** (virtual traffic lights) to prevent robot collisions and deadlocks in shared paths.

  Device Integration (NodeJS**,** ECS, Lambda, SQS, Dynamo, Postgres)
  - Build a device service that standardizes communication with elevators, automatic doors, and security gates, providing unified APIs and robust edge-case handling across robot vendors (Mitsubishi, Toshiba)

Dispatch Task Execution (Golang, Step Function, Lambda, Dynamo)

- Design and build a state-machine-based executor to decompose route plans into atomic actions, ensuring reliable delivery and error recovery.

Tech Stack: Golang, NodeJS, Postgres, Redis, Docker, AWS Services (Lambda, Step Functions, Glue, S3, Athena, IoT Analytics, ...)

- **ROBOVIZ**: Lead a team of 3 people. Build a visualization application from scratch; it has many visualizers, including video players for MJPEG-codec, charts, Point cloud displays, Google Maps, and a playback control to synchronize all visualizers by timestamp (NodeJS, React).
- **ROBOSIM**: Developed a dynamic form editor that enables users to define UI forms using XML standards.
- **CARIRO-MAP**: Build a graph-drawing tool to assist developers in creating simulation maps. It was created using the Konva library with React.
- **Z-RAPH**: Build a graph-based configuration tool from scratch. The core component is inherited from the react-diagrams library. The app validates all user input and dramatically reduces our system's setup time and error rate. This application can be used as a native app or in a browser (NodeJS, React, and Electron).

Research Assistant **Graphics and Computer Vision Lab - GVLab** **Aug 2019 - Sep 2020**

- Researched 3D medical image segmentation (CT blood vessels), proposed a variational U²Net3D model with deep supervision, improving the Dice score from 56% to 60% on the 3Dircadb dataset (PyTorch).
- Addressed class imbalance challenges in volumetric medical datasets using custom loss strategies.
- Supervisor: Ph.D. Le Thanh Sach, Ph.D. Nguyen Ho Man Rang.

Machine Learning Intern **CyberLogitec Vietnam Co.** **Jun 2019 - Aug 2019**

- Built a Chatbot for Airlines using Dialogflow by Google and integrated it into Messenger.

Personal Projects

Backend **Simple Bank** **Feb 2022 – Mar 2022**

- Self-learning how to completely design, develop, and deploy a backend service with key features: Create/Manage bank accounts, Record all balance changes to each account. Perform a money transfer between 2 accounts.
- Understand how to build a database consistently and reliably using transactions. Build a set of RESTful HTTP APIs using Gin, loading app configs, mocking DB with unit tests, handling errors, authenticating users, and securing the APIs with a JWT access token. Build and deploy with Docker.
- Repo: https://github.com/cs-tungthanh/Bank_Golang - Technique: Golang, PostgreSQL, Docker.

Backend **Microservice - Golang** **Mar 2022 – April 2022**

- Self-learning how a microservice is organized with Golang. Built several services: broker, logger, authentication, listener, and mail that can communicate with one another.
- Repo: https://github.com/cs-tungthanh/microservice-app - Techniques: Golang, PostgreSQL, Mongo, gRPC, RabbitMQ, Docker.

Education

Ho Chi Minh University of Technology **2016 – 2020**

- *Bachelor of Engineering - Computer Science*

Certification

AWS Certified Solutions Architect - Associate **Jul 2026 – Jul 2029**

- *Amazon Web Services (AWS)*